MILITARY SPECIFICATION
LANYARD, SAFETY, ASSEMBLY

1. SCOPE

1.1 This specification covers Lanyard, Safety, Assembly used on Flare, Aircraft, Parchute, MK24.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposals form a part of this specification to the extent specified herein.

SPECIFICATIONS

Federal
- PPP-B-636 - Box, Fiberboard

Military
- MIL-A-2550 - Ammunition and Special Weapons, General Specification for

STANDARDS

Federal
- Federal Test Methods Standard No. 151 - Metal; Test Methods

Military
- MIL-STD-105 - Sampling Procedures and Table for Inspection by Attributes
- MIL-STD-109 - Quality Assurance Terms and Definitions
- MIL-STD-129 - Marking for Shipment and Storage
- MIL-STD-1167 - Ammunition Data Cards

DRAWINGS

U.S. Army Munitions Command
- FC14425 - Lanyard, Safety, Assy
- FA32747 - Cable Energy Absorbing
- FB56641 - Poly Cell Matrix
- FC14446 - Poly Cell Envelope
- FA32870 - Sleeve, Swaging

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T00:51Z
Check the source to verify that this is the current version before use.

2.1 DRAWINGS (Cont'd)

FA32746 - Loop Disconnect
FA32745 - Loop Fastener
IEL-FC14425 - Inspection Equipment List for Safety Lanyard, for use on Flare Aircraft, Parachute, MK24

(Copies of specifications, standards and drawings required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 General.- Materials, parts and assembly shall comply with the requirements specified on the applicable drawings and referenced specification.

3.2 First article sample.- Requirements for the submission of first article sample by the contractor shall be as specified in the contract.

3.3 Material.- The contractor shall be responsible for selecting from the classification list contained in the raw material specification, the surface and physical condition of the raw material intended for use in manufacture.

3.4 Functional

3.4.1 The lanyard safety assembly shall function without mechanical failure (see 6.1).

3.4.2 The lanyard safety assembly under load (see 6.2) shall perform as follows:

3.4.2.1 The poly cell packaged lanyard safety assembly shall pay out automatically to its full length. It shall be capable of separating at the loop disconnect when a 30 to 50-lb tensile load at the rate of 3.5 inches per minute is applied to the loop fastener.

3.4.3 Temperature.- The lanyard safety assembly shall comply with all the functional requirements throughout the temperature range of minus 65 degrees to plus 125 degrees Fahrenheit (F).

3.5 Workmanship.- Requirements for workmanship are as specified on the applicable drawings, referenced specifications and the following:

3.5.1 Metal defects.- All components and assemblies shall be free from cracks, splits, cold shuts, inclusions, porosity, broken ends, frayed ends or any other defects.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T00:51Z
Check the source to verify that this is the current version before use.

3.5.2 Plastic defects.- All components and assemblies shall be free from blisters, bubbles, cracks, crazing, mottle, porosity, inadequate heat seal or any other defects.

3.5.3 Physical properties.- The physical properties specified on the applicable drawings and referenced specifications shall be uniform throughout each component.

3.5.4 Burr.- No part shall have a burr which might interfere with the assembly or function of the item or which might be injurious to personnel handling the item.

3.5.5 Foreign matter.- No part or assembly shall contain dirt, chips, grease, oil, rust, corrosion or other foreign matter.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specifications where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Quality Assurance terms and definitions.- Reference shall be made to MIL-STD-109 to define the quality assurance terms used.

4.1.2 Inspection.- Inspection shall be in accordance with MIL-A-2550, except as specified herein.

4.2 First article sample.

4.2.1 Initial production sample.- At the beginning of regular production, samples shall be submitted in accordance with contractual requirements and shall consist of 20 completed lanyard safety assemblies and 10 sets of unassembled components for physical and visual examination. The samples shall be manufactured using the same materials, equipment, processes and procedures as used in regular production. All parts and materials including packaging and packing, shall be the same as used in regular production and shall be obtained from the same source of supply.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T00:51Z
Check the source to verify that this is the current version before use.

4.2.1.1 Examination and test.- After inspection and provisional acceptance at source, the completed lanyard safety assemblies shall be inspected in accordance with 4.4. The unassembled components shall be inspected for all the requirements on the applicable drawings. These examinations and tests shall be performed at a Government laboratory or such other facility specified in the contract.

4.2.1.2 Initial production sample failure.- Failure of the sample to comply with the requirements of the drawings and specifications shall result in sample disapproval.

4.3 Inspection provisions.

4.3.1 Lot.

4.3.1.1 Submission of product.- The product shall be submitted in accordance with MIL-STD-105.

4.3.1.2 Size of lot.- A lot shall consist of an identifiable group of assembled units containing components produced by a homogeneous process.

4.3.2 Examination.- One-hundred percent examination shall be performed for all critical defects. Examination for major and minor defects shall be performed on a defect basis in accordance with the classification of defects contained herein, using the sampling plan and associated criteria in Table I and II shall be used. For an individual defect (single characteristic) the criteria in Table I shall be used. For total defects within a major or minor class the criteria in Table II shall be used. All nonconforming material shall be rejected. A four power (4X) magnification may be used in performing visual examination for such defects as burrs and foreign matter.

TABLE I

		Major		Minor	
Lot Size	Sample Size	Accept	Reject	Accept	Reject
1 - 300	100% Examination	-	-	-	-
301 - 1,300	110	1	2	2	3
1301 - 3,200	150	2	3	3	4
3201 - 8,000	225	3	4	4	5
8001 - 22,000	300	4	5	5	6

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T00:51Z
Check the source to verify that this is the current version before use.

4.3.2 (Cont'd)

TABLE II

Sample Size for MAJOR Defects	Total Allowable Defects per Sample Size			
	110	150	225	300
Number of defects in class				
2 - 3	1	2	3	4
4 - 7	2	3	4	5
8 - 19	3	4	5	7
20 - 43	4	5	8	10
Over 43	6	8	11	14
Sample Size for MINOR Defects	110	150	225	300
Number of defects in class				
2 - 3	2	3	4	5
4 - 7	3	4	5	7
8 - 16	4	5	8	10
17 - 48	6	8	11	14
Over 48	8	11	17	20

4.3.2.1 Assembly, drawing FC14425

Defect	Method of Inspection
CRITICAL	
None Defined	
MAJOR	
101. Metal defect	Visual
102. Improper assembly	Visual
103. Burr	Visual
104. Foreign matter	Visual

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T00:51Z
Check the source to verify that this is the current version before use.

4.3.2.1 (Cont'd)

Defect	Method of Inspection
MINOR	
201. Inside diameter loop fastener	Gage
202. Inside diameter loop disconnect	Gage
203. Length from cable to center of loop fastener	Gage
204. Length from cable to center of loop disconnect	Gage
205. Width of packaged assembly	Gage
206. Length of packaged assembly	Gage
207. Length of looped cable	Gage

4.3.2.2 Envelope, Drawing FC14446 covering a detail of drawing FC14425.

CRITICAL

None Defined

MAJOR	
101. Improper assembly	Visual
MINOR	
201. Length of heat seal at top of package	Gage
202. Location of heat seals	Gage

4.3.2.3 Matrix, Drawing FB 56641 covering a detail of drawing FC14425.

CRITICAL

None Defined

MAJOR

None Defined

MINOR	
201. Length	Gage
202. Width	Gage
203. Thickness	Gage

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T00:51Z
Check the source to verify that this is the current version before use.

4.3.2.4 Cable, Energy Absorbing, Drawing FA 32747 covering a detail of Drawing FC14425.

Defect	Method of Inspection
CRITICAL	
None Defined	
MAJOR	
101. Length	Gage
102. Diameter	Gage
103. Kink	Visual
104. Metal defect	Visual
MINOR	
201. Foreign matter	Visual
202. Burr	Visual

4.3.2.5 Loop Disconnect, Drawing FA32746 covering a detail of Drawing FC14425.

CRITICAL	
None Defined	
MAJOR	
101. Incorrect development	Visual
102. Burr	Visual
103. Metal defective	Visual
MINOR	
201. Inside diameter	Gage
202. Outside diameter	Gage
203. Inside diameter of hole	Gage
204. Depth of hole	Gage
205. Overall length	Gage

4.3.2.6 Loop fastener, Drawing FA32745, covering a detail of drawing FC14425.

CRITICAL	
None Defined	

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T00:51Z
Check the source to verify that this is the current version before use.

4.3.2.6 (Cont'd)

Defect	Method of Inspection
MAJOR	
101. Incorrect development	Visual
102. Outside diameter	Gage
103. Burr	Visual
104. Metal defective	Visual
MINOR	
201. Inside diameter	Gage
202. Overall length	Gage

4.3.2.7 Sleeve, Swaging, Drawing FA32870 covering a detail of Drawing FC14425.

CRITICAL

None Defined

MAJOR	
101. Metal defect	Visual
102. Burr	Visual
MINOR	

None Defined

4.3.2.8 Packing in Fiberboard Box

CRITICAL

None Defined

MAJOR

None Defined

MINOR	
201. Contents loose	Manual
202. Box cut or damaged	Visual
203. Number of units incorrect	Visual
204. Marking missing, incorrect or illegible	Visual

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T00:51Z
Check the source to verify that this is the current version before use.

4.3.2.9 Process control.- Periodic unscheduled monitoring shall be conducted for the following process operations to determine compliance with the applicable drawing:

Operation	Drawing
Heat seal	FC14425
Stress relieve	FA32745

4.3.3 Functional testing.- The following test shall be performed in accordance with the provisions of 4.4.

4.3.3.1 Functional.- Forty-five completed lanyard safety assemblies shall be tested in accordance with 3.4 under the following conditions:

Sample size	Temperature
15	- 60° to - 65°F
15	+ 65° to + 75°F
15	+120° to +125°F

Failure of any assembly to comply shall be cause for rejection of the lot of assemblies.

4.4 Test methods and procedures.

4.4.1 Functional.- The completed lanyard safety assembly shall be tested using a simulated load in a free fall drop, extending the lanyard to its full length with sufficient pull to cause the loop disconnect, to function at the static line end with the required pull within the specified load rate.

4.4.2 Temperature conditioning.- The completed lanyard safety assemblies shall be stored in a temperature conditioning chamber at the temperatures specified in 4.3.3.1 to reach the required test temperatures before testing.

5. PREPARATION FOR DELIVERY

5.1 Packing.- One hundred each lanyard safety assemblies shall be packed and shipped in fiberboard boxes in accordance with PPP-B-636.

5.2 Marking.- Shipments shall be marked in conformance with MIL-STD-129.

5.3 Data card.- Data card information shall be as specified in MIL-STD-1167.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T00:51Z
Check the source to verify that this is the current version before use.

6. NOTES

6.1 A "mechanical failure" is defined as any deformation or breakage of a part, the occurrence of which is other than a design function.

6.2 Load.- The expected performance of the lanyard safety assembly, under conditions simulating those existing in actual use shall be referred to as load conditions in this specification.

6.3 Ordering data.- Procurement documents should specify the title, number and date of this specification.

Custodian:

Army - MU

Preparing activity:

Army - MU

Project No. 1370-A300

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T00:51Z
Check the source to verify that this is the current version before use.